

Finding Nemo, if Nemo was Electrically Neutral Hydrogen in the Milky Way Galaxy

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ABSTRACT

1. Introduction

One of the hyperfine lines produced by the partial decay of electrically neutral hydrogen results in the spontaneous spin flip of the hydrogen's electron. Since neutral hydrogen is quite stable, this phenomenon takes approximately ten million years. This specific transition causes energy to be released as the electron moves to a lower energy level, which corresponds to a photon wavelength of about 21-cm and a frequency of 1420.405 MHz. Hence, this transition is often referred to as the 21-cm line when observed. The 21-cm line is commonly utilized in radio astronomy for analyzing rotation curves in galaxies, mapping galaxy structures, and uncovering early structures in the universe because of its ability to travel through clouds of gas that are opaque to other wavelengths of light. This experiment will locate, capture, and analyze the 21-cm line and utilize system calibrations to identify the spiral arms in the Milky Way Galaxy using the Big Horn radio telescope, located at Campbell Hall at the University of California, Berkeley.

In §2 we introduce the techniques used in this paper. In §3.1 we describe the various methods and experiments performed, which are analyzed in §4 and we state our conclusions in §6.

2. Theoretical Background

In order to determine whether the neutral hydrogen spectrum, commonly known as the 21-cm line, was observed in the spiral arms of the Milky Way, the Big Horn radio frequency telescope and receiver system was strategically calibrated to capture spectra at various locations on the sky. Cassiopeia was the constellation chosen to point at because

of its in a convenient location in the sky relative to the observation site, and because it is in the direction of one of the spiral arms of the Milk Way. Thus, we expect to observe the 21-cm line while pointing at Cassiopeia if the equipment is correctly calibrated. The maximum observable frequency must be equal to or less than the observed frequency, as given in terms of the sampling frequency. Since the HI line is emitted at 1420.405 MHz, the correct sampling frequency can be found using the following equation:

$$f_{\max} \leq f_{\text{nyquist}} = \frac{f_{\text{sample}}}{2} \quad (1)$$

The equation above is commonly referred to as the Nyquist Criterion, and it dictates the zones where a signal can be observed without aliasing. The software defined radio (SDR) used throughout the experiment consists of an onboard tuner and a local oscillator (LO) that allow us to control the sampling rate and collect raw data. By setting the LO to a given center frequency we were able to capture higher frequency signals, as demonstrated by the following equation:

$$f_{\text{observed}} = f_{\text{signal}} - f_{\text{center}} \quad (2)$$

Therefore, in order to observe the 1420.405 MHz signal with a small spectral band, we offset the center frequency by a few hundred kilohertz by setting the center frequency to about 1420.395 MHz. This avoids the spike in the power spectrum that usually occurs near the zero bin while remaining under the SDR’s sampling rate limit of 3.2 MHz. Since the HI line is quite dim, it is important to optimize and correctly capture the data from the receiver system; methods of ensuring high quality data include checking for signal clipping (removes high frequency signals) and a receiver continuity with a random noise test. Both methods can be confirmed by fitting the respective power distributions to a Gaussian model, which confirms that the high frequency are not clipped, and that taking data without a target produces random noise from radio frequency interference. In order to locate the 21-cm line, three different rotation matrices must be utilized to convert from galactic coordinates given in latitude and logitude. The following series of matrices can be multiplied to convert from galactic, to equatorial, then finally local coordinates in their respective orders. Note the matrix further to the right is inverse because it is normally written for equatorial to galactic conversions, but for our experiment we use the inverse to do the reverse operation.

$$\begin{bmatrix} -\sin(\phi) & 0 & \cos(\phi) \\ 0 & -1 & 0 \\ \cos(\phi) & 0 & \sin(\phi) \end{bmatrix} \begin{bmatrix} \cos(\text{LST}) & \sin(\text{LST}) & 0 \\ \sin(\text{LST}) & -\cos(\text{LST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} \left(\begin{bmatrix} -0.054876 & -0.873437 & -0.483835 \\ 0.494109 & -0.444830 & 0.746982 \\ -0.867666 & -0.198076 & 0.455984 \end{bmatrix} \right)^{-1} \quad (3)$$

Once the correct coordinates for the desired object are found from the above coordinate conversions, the telescope can be pointed accurately to capture the HI line. After observing the line, the telescope can be pointed to zenith or a different part of the sky to ensure the line is strongest when the telescope is being intentionally pointed at a sources that is expected to have a stronger HI line. The resulting power spectra can be fit to multiple Gaussian curves to confirm that their respective peaks are at the expected frequency. This is due to the fact that particles in a cloud of gas follow the Maxwell-Boltzmann distribution principle: a special case of a Gaussian model for the three dimensional motion of a gas particle. Additionally, the observed spectra must be shifted in order to account for the Doppler effect caused by Earth's motion relative to the Milky Way. This shift can be found using the following equation:

$$f_{\text{observed}} = \left(\frac{v \pm v_0}{v \mp v_s} \right) f_{\text{emitted}} \quad (4)$$

The variables can be defined as follows: v is the wave speed in a medium, v_0 is the speed of the observer relative to the source, and v_s is the source velocity. Since the source in this case is a constellation, the source velocity would be zero. As a first step towards observing the HI line with all of the above information taken into consideration, the initial experiment towards consisted of confirming the Big Horn radio telescope and receiver system functions as expected.

3. Methods and Experiments

The Realtek Software Defined Radio (RTL2832U SDR, or SDR for short) used throughout the experiment consists of an onboard mixer that was bypassed to allow us to directly sample data. The RTL SDR consists of two chips; the first is the Rafael Micro R820T2 chip which serves as a radio frequency (RF) tuner that filters, amplifies, mixes the signal with an LO, then uses analog low-pass filtering to output analog I and Q signals. Those signals are then fed into the second chip, the RTL2832U Demodulator / ADC, which decimates and downconverts the signal to output digital I/Q samples through a USB port. In the first experiment, we set out to test if the system was working as we expected it to, so we

sent an injected signal up to the telescope and tested if we could sample at correct rate for frequencies near 1420 MHz. Once this was confirmed, we were able to move onto testing the baseline input from the Big Horn by pointing the aperture to zenith and taking data samples. Only after we were confident that we were capturing data from the sky, we were able to calculate the correct local coordinates to point to $(120^\circ, 0)$ in the galactic plane. These coordinates correlate with the constellation Cassiopeia which is known to be in the direction of one of the spiral arms in the Milky Way to an observer on Earth. Consequently, pointing at Cassiopeia should indicate a stronger HI line as hydrogen is present as clouds of gas in the spiral arms of our galaxy. Only after an undeniable spectra of the 21-cm line was taken did we move onto calibrating the system by accounting for gain, impedance, and characterizing the cables and antenna in our system.

3.1. Experiment 1: Configure Setup

The Big Horn system is equipped with an antenna inside the horn through which an injected signal can be sent and controlled from the lab. Since the ultimate goal of this report is to accurately capture the 21-cm line, we initially injected a signal at 1420.405 MHz, the same as the 21-cm line, and set the amplitude to -80 dBm on the wave generator. The purpose for having an amplitude so low is because the HI line itself is not very strong, so we found it necessary to test the sensitivity of the telescope. Unfortunately, due to the low amplitude of the signal the noise of the system overcame the signal until about 20 dBm where the signal finally came through, as is shown in the figure below.

Since the receiver was responding as we expected to the injected signal, we then tested to system by taking a spectrum of the sky with the telescope aperture pointing to zenith. Since there was no specific signal injected or targeted, the expected result from this was random noise. In order to confirm our theory, we fit the data collected from zenith to a Gaussian curve, as truly random noise should result Gaussian distribution as seen in Fig. 2.

3.2. Experiment 2: 21-cm Line Spectra

Utilizing the matrices shown in (3), we were able to point to Cassiopeia at galactic coordinates $(120^\circ, 0)$ after we found the local coordinates at approximately $(34^\circ, 36^\circ)$ while observing from Campbell Hall, Berkeley, CA. Once the HI line was clearly captured, we tested our confidence by observing the HI line in different parts of our bandpass filter, otherwise known as frequency switching. Since the HI line is a low signal, numerous data blocks were

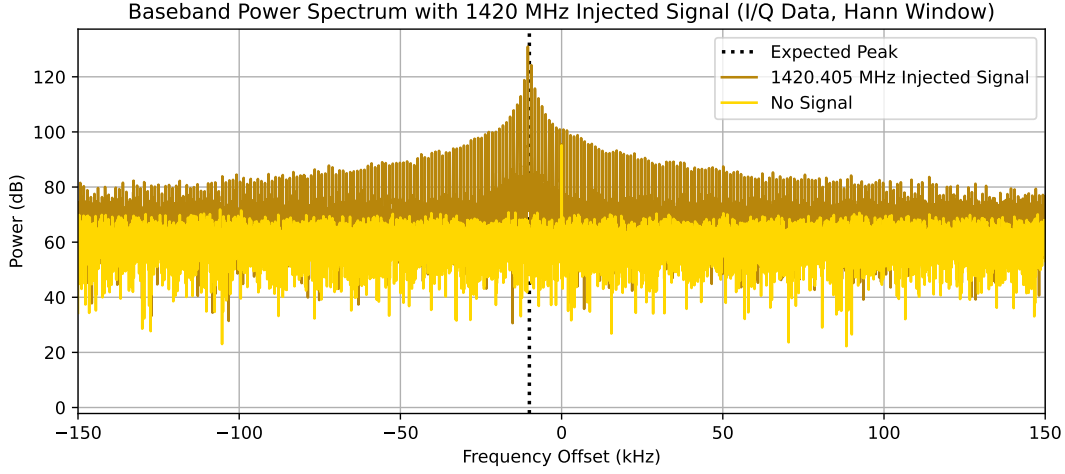


Fig. 1.— This plot displays the power spectrum observed when an injected signal at 1420 MHz was injected into the receiver system. The brighter yellow strip along the bottom of the graph is the power spectrum from the horn with the same injected signal but the amplitude was at 0 dBm so it was not visible through the noise. On the other hand, the darker yellow spectrum with the large peak is with the signal with an amplitude of 20 dBm. The dotted black line is where we expected the peak to be based off our calculations from eq. (2); the local oscillator was set to 120.395 MHz and the injected signal was set to 1420.405 MHz, giving a 10 kHz offset which is what is observed above. As the predicted and the experimental data match, we were able to conclude that the horn system was receiving signals correctly and moved onto the next experiment.

averaged and smoothed over in order to produce a distinct line with minimal noise. As shown in Fig. 3, the result of using frequency switching is two smoothed power spectra with peaks on opposite sides of the bandpass that confirm the 21-cm line was properly targeted and observed.

3.3. Experiment 3: Hardware and Temperature Calibration

Finish this section with a description of the data you collected (number of samples, time interval, sampling frequency, etc.)

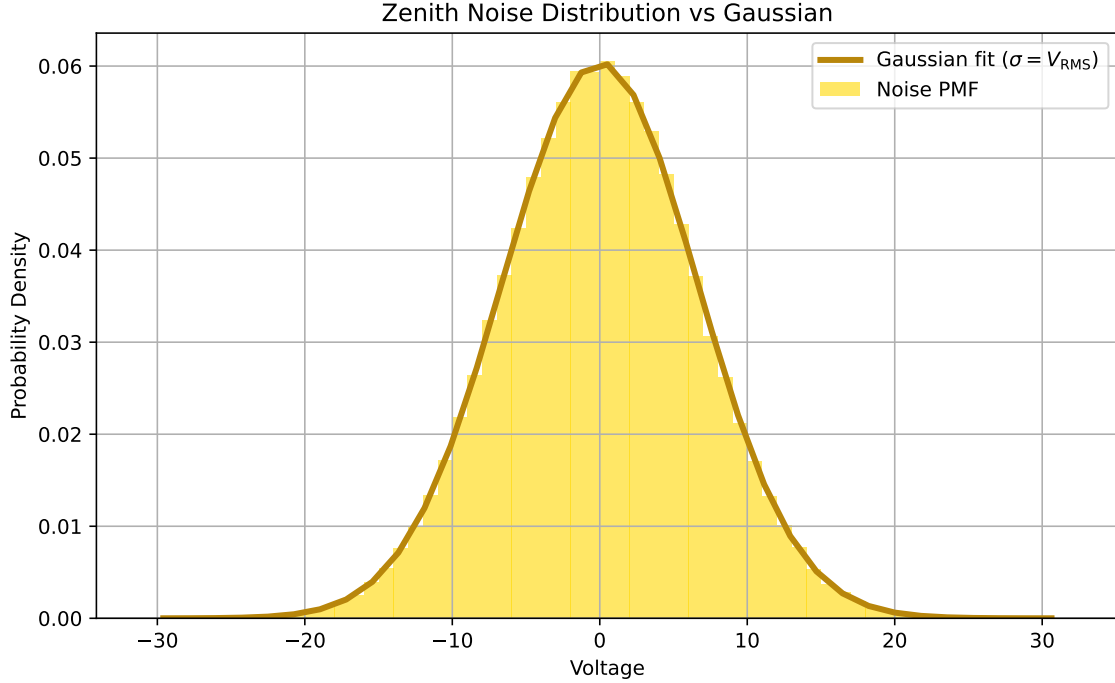


Fig. 2.— Above shows the raw data collected from pointing the telescope to zenith, plotted using a histogram function from the python package numpy, and then fitted to a Gaussian. The data fits the Gaussian well with a peak around 0.06 and a RMSE of $6.45\text{e-}03$. Further indicating that the signals received from pointing to zenith are essentially random noise, which is commonly known as radio frequency interference (RFI) when trying to observe other objection intentionally. Once again, our expectations were confirmed with the observation, allowing us to move onto targeting specific a specific object to observe the 21-cm line.

4. Analysis

Present findings with error. Use linear non-linear least-squares methods to fit models estimate error on the following variables:

HI line peak doppler corrected, system temp, system gain, signal delay & power loss, aperture efficiency

5. Results

Reiterate findings and how the line up with what we expected. Also reiterate how our assumptions from 2 could have effected the data & findings for the experiment.



Fig. 3.— The On and off spectra above demonstrate frequency switching by deliberately changing the LO value to move the HI peak from the upper left to the lower right of the band pass. Note the on spectra has a significant dip where the off spectra has a peak. Using this data for further analysis may carry over this dip, which may have been an artifact of imaging different parts of the sky due to not moving the telescope frequently enough between observations.

Measurements with error bars are key results that you should highlight here. Highlight these results in your abstract, too.

6. Conclusion

We set out to observe and confirm the presence of the 21-cm line in the spiral arms of Milky Way Galaxy, and were able to conclude with (insert level of error) that the peak at (insert Kelvin) at approximately 1420.405 MHz was due to the presence of gaseous neutral hydrogen. Things to cover: confidence in data collection based on calibrations (cable delay, aperture efficiency, system gain/temp/noise); things that could be improved if the experiment was replicated: smaller data blocks (better storage), more precise pointing (or at least more rigor on getting the error from the pointing technique), not collecting data without repointing the telescope at least every 20 minutes, and getting better calculation on the power loss / time delay from the roof.

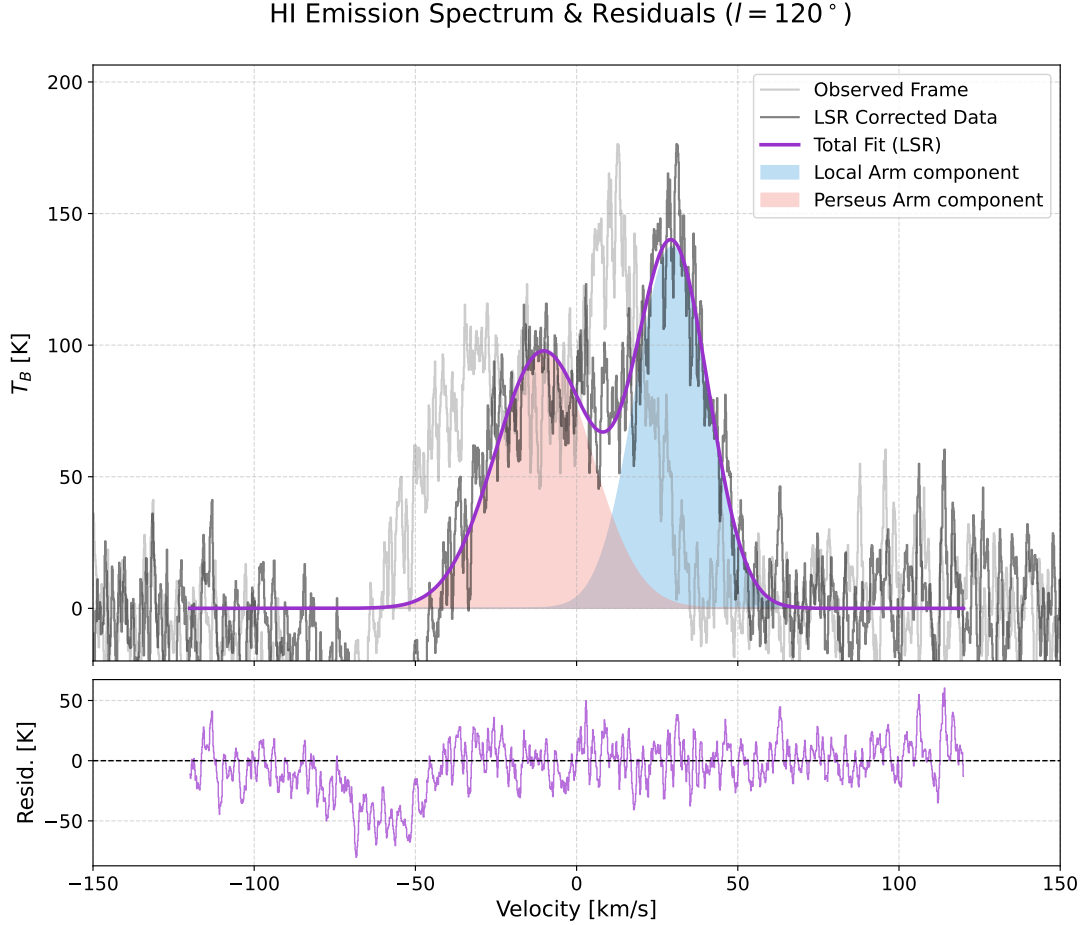


Fig. 4.— Above is the temperature spectrum before and after the Local Standard of Rest was corrected for. Additionally, to verify that the peaks shown are from the 21-cm line they were fitted to two individual Gaussian curves, which is expected based on the Maxwell-Boltzmann distribution of an ideal gas. Note the matching dip in the emission spectrum and the residual, which aligns with the dip in the ON spectra from Fig. 3.

Acknowledgements

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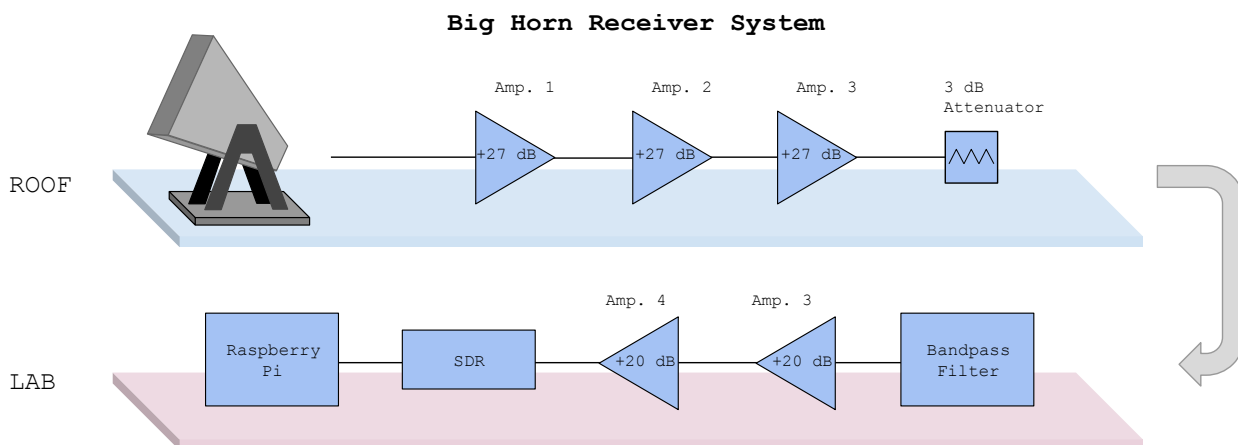


Fig. 5.— The block diagram above shows the signal chain from the Big Horn antenna located on the roof, to the data taking system in the lab. Overall, the signal must be picked up by the antenna, go through three amplifiers with $+27$ dB gain, then a 3 dB attenuator on the roof. The signal is then transported to the lab through a cable, which delivers data through a bandpass filter, two more amplifiers with $+20$ dB gain, until finally data is collected with the SDR and downloaded to the Raspberry Pi.

following python packages were used throughout the entirety of this lab: `numpy`, `ugradio`, `matplotlib.pyplot`, and `astropy`. AI was used exclusively as aid for part of the production and debugging of plotting material and minor grammatical errors throughout the report.